UniDec: A Unified Factor-Graph-Based Decoder Fully Compatible With 5G NR LDPC/Polar Codes

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***Abstract*—In comparison to 4G, 5G wireless needs to support a broader range of applications. Therefore, both low-density parity-check (LDPC) codes and polar codes have been standardized by 5G new radio (NR) to fulfill the requirements of data channel and control channel, respectively. Usually, LDPC/polar decodings are implemented by separate hardware, leading to low area efficiency. Though decoders which can handle both codes have been proposed, how to compromise between throughput and efficiency has always been a persistent dilemma due to the absence of a unified and smooth integration methodology. To this end, by fully utilizing the common parts of graph-theoretic algorithms for both codes, this paper presents a unified decoder (UniDec) which is fully compatible with 5G NR LDPC/polar codes. This UniDec enables three key approaches: *1) unified processing nodes for both codes*, *2) configurable permutation networks with multi-parallelism*, and *3) flexible scheduling for 5G NR parameter configuration*, guaranteeing both high data throughput and area efficiency. Implemented in 40nm CMOS, the UniDec attains a maximum of 33.64× throughput and 5.98× area efficiency compared to its multi-mode counterparts. Even compared with the state-of-the-art (SOA) dedicated ones, the UniDec still maintains a competitive edge in terms of throughput, energy, and area efficiency. It is noted that this methodology can be generalized to other factor-graph based signal processing algorithms.**

***Index Terms*—Graph-based architecture, LDPC codes, polar codes, message passing, unified decoder.**

I. Introduction

IN CONTRAST to 4G limited for a single application scenario, 5G wireless needs to support three major application scenarios while achieving satisfactory performance [1]. Aiming to support diverse application scenarios [2], 5G must be capable of delivering differentiated and customized services. In 5G, baseband processing is one of the components that integrate most core technologies. Effectively integrating these key technologies to provide diverse services will be essential for the future development of baseband processing. However, the separate technologies are usually integrated roughly for implementation to meet various demands, resulting in numerous challenges such as high costs, high power consumption, and limited flexibility [3]. Therefore, a unified integration methodology for baseband processing algorithms should be fully explored to enhance the efficiency of hardware implementations.

In the realm of baseband processing technologies for 5G, channel coding is crucial for achieving capacity in transmission. Both low-density party-check (LDPC) codes [4] and polar codes [5] have been standardized by 5G new radio (5G NR) in enhanced mobile broadband (eMBB) scenario [6]. They are employed respectively as data channel and control channel coding to fulfill communication requirements. Numerous algorithms [7], [8], [9] related to these two codes have attracted great attentions and experienced significant development [10], [11], [12], [13]. In practical application, the decoding algorithms for LDPC/polar codes have been implemented in various hardware architectures [14], [15], [16], [17] to improve the throughput, efficiency and flexibility. An unrolled parallel LDPC decoder [14] is proposed to achieve ultrahigh throughput but has limited configurability. To support multiple code rates and block lengths, the reconfigurable circular-shift networks [18], [19] are employed to handle various lifting values with different permutation constraints. In [20], an area efficient LDPC decoder [20] is proposed to support all the lifting sizes specified in the 5G NR standard. Its decoding throughput is greater than 20Gbps. To reduce the hardware complexity further, a novel check-node LDPC decoder [21] is presented, offering improved area-throughput performance. The polar decoder [22] is designed with optimal parameters to make trade-offs between the latency and area overheads. The aforementioned dedicated decoders are constrained to a single decoding algorithm. To optimize both throughput and efficiency for multi-codes, multi-mode decoders are considered to integrate LDPC/polar decoding implementations.

Consequently, several studies have focused on multi-mode decoders [23], [24], [25], [26], [27] to provide the diverse communication services. Guessing Random Additive Noise Decoding (GRAND) [28] is a universal decoding scheme for any linear block codes. The multi-mode GRAND decoders [29], [30] allow the superior error-correction performance for short-length codewords. However, their excessive complexity

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with long-length codewords prevents them from achieving full compatibility with 5G NR standard. In [23], a merged factor graph scheme is proposed for belief propagation (BP) decoding. Based on BP decoding, a joint LDPC-polar decoding structure is presented in [24]. To this end, the first LDPC-polar decoder is introduced in [26] for the 5G NR standard. The decoder combines the decoding algorithms by the add-comparison-add calculation pattern. However, decoding mode and pattern configurations restrict throughput and efficiency in various scenarios. The dual-mode 5G decoder [31] supports the lifting size-128 in LDPC mode and the length-1024 in polar mode, but lacks configurability for various scenarios. Therefore, there is still a need for a unified and smooth integration methodology to develop an efficient hardware architecture for multi-mode decoding.

In this paper, we propose a unified factor-graph based decoder (UniDec) that is fully compatible with 5G NR LDPC/polar codes. Based on the graph theory in [32], the UniDec utilizes the common parts of graph-theoretic algorithms for both codes to reduce the implementation complexity. By describing the common aspects using the hardware formula representation [33], we propose a unified integration methodology to derive the factor-graph based architecture for 5G NR LDPC/polar codes. In this architecture, the unified processing nodes and configurable permutation networks are proposed to improve throughput and efficiency. Besides, the flexible scheduling for both codes is supported through 5G NR parameter configuration to ensure 5G NR compatible decoding.

Contributions and Paper Outline:

The contributions of this work are summarized as follows.

- Based on our unified integrated methodology, we utilize the hardware formula representation to derive the UniDec for LDPC/polar codes. The UniDec consists of core decoding logic units and extrinsic memory units. The core decoding logic units include unified processing nodes and configurable permutation networks, aiming to enhance throughput and hardware efficiency. The extrinsic memory units serve to store intermediate soft messages and control instructions for flexible scheduling.
- In core decoding logic units, the unified processing nodes are proposed by extracting the common operators for LDPC/polar decoding. This allows these unified processing nodes to share decoding logic and reconfigure with different functions for both codes, reducing hardware complexity. Configurable permutation networks are introduced to achieve the interconnection of unified processing nodes. To maximize throughput and efficiency of the network, multi-parallelism control schemes are also proposed to support multi-frame processing for both codes.
- In extrinsic memory units, the data information memory is shared to store intermediate soft messages for memory saving. The control information memory is dedicated to storing control instructions for 5G NR parameter configuration. The instructions are constructed by extracting code parameters such as code lengths, code rates, and memory access conflicts. Based on these 5G NR parameters, the decoder follows the intrinsic decoding rules for flexible scheduling, ensuring full compatibility with 5G NR standards for LDPC/polar decoding.

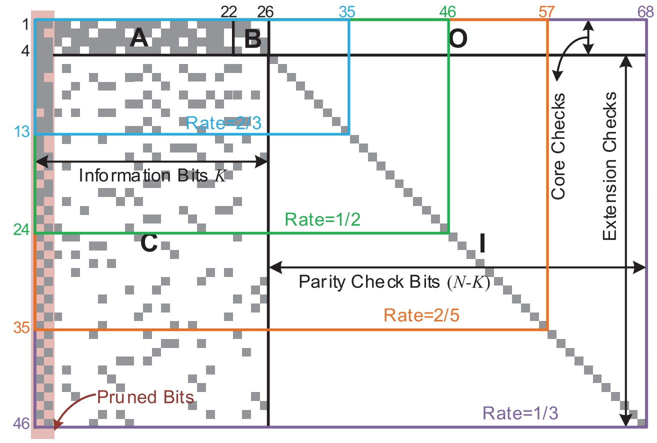


Fig. 1. Schematic representation of BG1 in 5G NR standard.

Based on the aforementioned methods, the proposed architecture is implemented in an area of 1.64 mm^2 at 500 MHz based on 40 nm CMOS technology. It can deliver peak throughputs of 21.87 Gb/s and 10.24 Gb/s in LDPC and polar modes, respectively. Compared with the state-of-the-art (SOA) multi-mode decoder [26], our decoder attains a maximum throughput of 33.64$\times$ and an area efficiency of 5.98$\times$, respectively.

The rest of this paper is organized as follows. Section II reviews the basic structures of 5G NR LDPC/polar codes and hardware formula representation. In Section III, we derive the unified factor graph based architecture with the hardware formula representation. The hardware design details of the UniDec are provided in Section IV. The hardware results are provided to compare with the SOA multi-mode and dedicated decoders in Section V. Section VI draws the conclusions.

II. Preliminaries

A. 5G NR LDPC/Polar Codes

In 5G eMBB scenarios, LDPC codes are utilized as the data channel codes, employing a quasi-cyclic structure. Quasi-cyclic LDPC (QC-LDPC) codes are represented by an $m_b \times n_b$ base graph, where each element in the base graph corresponds to a $Z \times Z$ submatrix. This submatrix can either be a zero matrix or a cyclically shifted identity matrix, determined by the zero-one elements and shifted values in the base graph. Consequently, base graphs can accommodate various information bit lengths K and code rates $R = \frac{K}{N_b}$ by adjusting the parameter Z for an $M_b \times N_b$ parity-check matrix (PCM) $\mathbf{H}$, where $M_b = Z \cdot m_b$ and $N_b = Z \cdot n_b$.

To satisfy the diverse communication requirements across various scenarios, 5G NR LDPC codes employ two distinct types of base graphs, BG1 and BG2. The BG1 structure is shown in Fig. 1 with dimensions 46×68, supporting code rates from $1/3$ to $8/9$ and the maximum information bit length 8848. The BG2 has 42 rows and 52 columns that supports

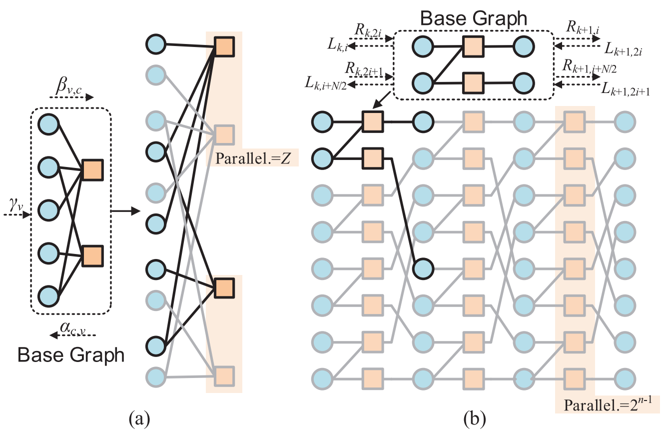


Fig. 2. BP decoding factor graphs derived from the base graphs of (a) QC-LDPC codes and (b) polar codes.

code rates from 1/5 to 2/3 and the maximum information bit length 2540. To ensure the scalability, there are 51 lifting sizes Z ranging from 2 to 384 to generate PCMs [34], providing various ranges of information bit lengths and code rates to meet communication demands.

Polar codes are the first codes proven to achieve channel capacity $I(W)$ in a binary-input discrete memoryless channel W. They synthesize N independent copies of channels W to be a set of polarized channels $W_N^{(i)}, 0 \le i < N$. In practice, the bit channels whose $I(W_N^i)$ approach 1 are selected to construct information set $\mathcal{A}$ with a size of K and the remaining bit channels as frozen bits transmit fixed bits, usually value 0. The encoding processing can be expressed as $\boldsymbol{x} = \boldsymbol{u}\mathbf{G}$, where $\mathbf{G} = \mathbf{F}^{\otimes n}$ is the generator matrix, $\mathbf{F}^{\otimes n}$ is the n-th Kronecker product of $\mathbf{F} = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$, $n = \log_2 N$. And the notations $\boldsymbol{x}$ and $\boldsymbol{u}$ denote the encoded and input vectors.

Polar codes have been selected for encoding control information in the 5G eMBB scenario. These codes provide a code length range of 2^n, where $7 \le n \le 9$ for downlink. Considering the significant difference in code length range compared to 5G NR LDPC codes, the decoding architecture design faces challenges in achieving both flexibility and efficiency for both codes simultaneously. Therefore, it is crucial to introduce a hardware architecture based on a unified decoding algorithm that effectively balances generality and specificity.

B. BP Decoding for 5G NR LDPC/Polar Codes

BP decoding is a message-passing algorithm employed for both LDPC/polar codes based on factor graphs. The factor graph consists of check nodes (CNs) and variable nodes (VNs), and the associated messages are updated iteratively through permutations between CNs and VNs.

For LDPC codes, the $\alpha_{c,v}$ and $\beta_{v,c}$ denote the log-likelihood ratio (LLR) messages passed from the CN c to the VN v and from the VN v to the CN c, respectively. And γ_v denotes the a-posteriori-probability LLR (AP-LLR) messages of the VN v. The expression $\mathcal{V}_c/\mathcal{C}_v$ represents all VNs/CNs connected to CN c/VN v, and the degree d_c/d_v denotes the size of $\mathcal{V}_c/\mathcal{C}_v$. The CN and VN message processing as Fig. 2(a) is presented as follows:

1) VN update: In the t-th iteration, $\beta^{(t)}_{v,c}$ is calculated as

$$\beta^{(t)}_{v,c} = \gamma^{(t)}_v - \alpha^{(t-1)}_{c,v}. \tag{1}$$

2) CN update: In the t-th iteration, $\alpha^{(t)}_{c,v}$ is calculated as

$$\alpha^{(t)}_{c,v} = \varphi(\beta^{(t)}_{v,c}). \tag{2}$$

3) AP-LLR update: In the t-th iteration, $\gamma^{(t)}_v$ is calculated as

$$\gamma^{(t)}_v = \alpha^{(t)}_{c,v} + \beta^{(t)}_{v,c}. \tag{3}$$

To perform an approximate calculation, offset min-sum (OMS) algorithm [35] are applied for $\varphi(\beta^{(t)}_{v,c})$ function as

$$\prod_{v'\in\mathcal{V}_c\setminus v} \mathrm{sgn}(\beta^{(t)}_{v',c}) \max\Big(\min_{v'\in\mathcal{V}_c\setminus v}(|\beta^{(t)}_{v',c}| - \theta), 0\Big), \tag{4}$$

where the notation sgn() denotes signum function, and θ is an offset factor. The $\alpha^{(0)}_{c,v}$ and $\beta^{(0)}_{v,c}$ are initialized with zero.

For polar codes, an n-stage factor graph contains $(n+1)\times N$ nodes, and each stage includes $\frac{N}{2}$ PEs. Fig. 2(b) shows the factor graph for polar codes, $N = 8$. Two types of directional messages $L_{k,i}/R_{k,i}$ are propagated over the factor graph, where k and i denote the stage and bit indices. The $\boldsymbol{L}/\boldsymbol{R}$ messages processing are presented as follows:

$$\begin{cases} L_{k,2j} = \phi\left(L_{k+1,j}, R_{k,2j+1} + L_{k+1,j+\frac{N}{2}}\right), \\ L_{k,2j+1} = \phi(L_{k+1,j}, R_{k,2j}) + L_{k+1,j+\frac{N}{2}}, \\ R_{k+1,j} = \phi\left(R_{k,2j}, R_{k,2j+1} + L_{k+1,j+\frac{N}{2}}\right), \\ R_{k+1,j+\frac{N}{2}} = \phi(R_{k,2j}, L_{k+1,j}) + R_{k,2j+1}, \end{cases} \tag{5}$$

where j denotes the j-th PE at one stage. The $\phi(a, b)$ function is the OMS function [36] that expressed as

$$\mathrm{sgn}(a)\mathrm{sgn}(b)\max(\min(|a|, |b|) - \theta,\ 0). \tag{6}$$

In summary, it is feasible to achieve both LDPC/polar decoding through BP algorithm within graphical models. However, the graphical structures derived from different base graphs in Fig. 2 encounter obstacles in transitioning from algorithmic models to unified hardware implementations. Hence, it is necessary to provide a hardware description for various architectures. Subsequently, extracting and summarizing common characteristics from this description can lead to a unified hardware architecture. In the following section, fundamental concepts of hardware description will be introduced before presenting our proposed unified hardware architecture.

C. Formula-Based Module Design and Connection

The hardware signal processing language (HSPL) [33] is proposed to provide a formal representation for complicated systems. The fundamental representation of the data path for any module can be denoted as:

$$\mathbf{O} = \mathbf{I} \rightarrow \mathbb{M}, \tag{7}$$

where $\mathbf{I}$ and $\mathbf{O}$ refer to the input vector and output vector of the module $\mathbb{M}$.

Considering different data paths and module connections, Fig. 3 illustrates different formula representations in a cascade or parallel connection. Fig. 3(a)-(b) show the data path passes

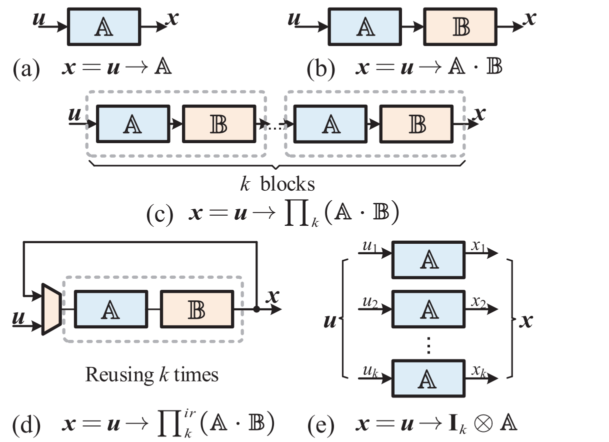


Fig. 3. Formula hardware architecture representations in HSPL (a) $\boldsymbol{x} = \boldsymbol{u} \rightarrow \mathbb{A}$ (b) $\boldsymbol{x} = \boldsymbol{u} \rightarrow \mathbb{A} \cdot \mathbb{B}$ (c) $\boldsymbol{x} = \boldsymbol{u} \rightarrow \prod_k (\mathbb{A} \cdot \mathbb{B})$ (d) $\boldsymbol{x} = \boldsymbol{u} \rightarrow \prod_k^{ir} (\mathbb{A} \cdot \mathbb{B})$ (e) $\boldsymbol{x} = \boldsymbol{u} \rightarrow \mathbf{I}_k \otimes \mathbb{A}$.

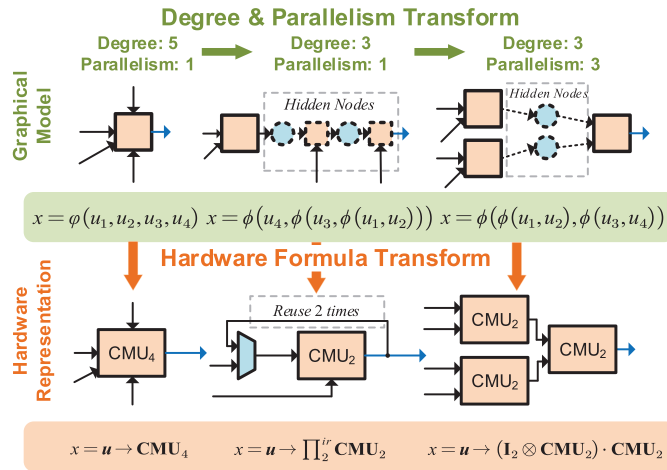


Fig. 4. Degree and parallelism transformation for the codes on graphs.

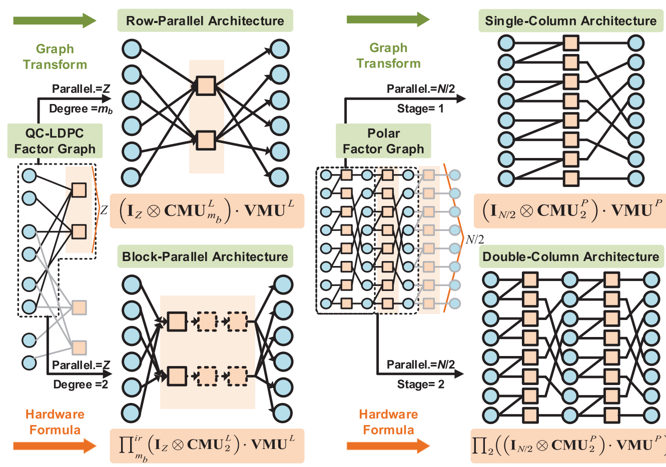


Fig. 5. Hardware representations for the existing decoding architectures for LDPC and polar codes.

a module $\mathbb{A}$, module $\mathbb{A}$ and module $\mathbb{B}$ in series, respectively, where $\boldsymbol{u}$ and $\boldsymbol{x}$ denote input and output, the interpunct symbol $\cdot$ denotes the cascade connection. The same type of modules connected in sequence are shown in Fig. 3(c), where the product symbol $\prod_k$ replaces the interpunct symbol to denote the cascade connection with k modules. In Fig. 3(d), the representation of iteratively reusing k times can achieve the same function as that in Fig. 3(c), where the symbol $\prod_k^{ir}$ refers to the iterative data path with k times. Finally, the symbol $(\mathrm{I}_k \otimes)$ indicates the parallel connection for k placed modules, and the corresponding representation is illustrated in Fig. 3(e).

Therefore, the HSPL provides a method to describe the algorithmic model at the hardware level. Based on the aforementioned hardware representation, we can articulate the hardware formulas for different graph structure. By extracting common factors from these hardware formula, we can achieve an efficient and unified hardware architecture for both LDPC and polar codes on the factor graph.

III. Hardware Formula Representation for Codes on Graphs

BP algorithms can uniformly perform decoding for codes on graph models at the algorithm level [32]. Inspired by this, we extract and analyze the common features of factor graphs based on hardware formula representation. Subsequently, a unified integrated methodology can be proposed to derive the factor-graph based architecture at the hardware level.

In this section, we consider the hardware formula representation for the nodes in the factor graph. Specifically, we focus on two key attributes of these nodes: degree and parallelism, both of which need to be expressed in terms of hardware representations. The degree reflects the hardware complexity of the nodes, and the parallelism shows the number of instantiations of the nodes. By breaking down high degree nodes into multiple nodes with lower degrees, we can standardize the diverse factor graph structures of LDPC codes and polar codes at the hardware level.

The equations of (4) and (6) represent message-passing decoding for nodes of different degrees in the factor graph. While the factor graph of LDPC codes primarily consist of high degree-d, $d > 3$ nodes, there are only degree-3 and 2 nodes in polar codes. Therefore, we present the degree and parallelism transformations to unify the graph structures of these two codes in the hardware representation. Fig. 4 shows the degree and parallelism transformations of a degree-5 CN, and the corresponding hardware formula representations as an example, where the CMU_i represents the CN function unit with i inputs.

In Fig. 4, we present the nodes with different degrees and parallelism in hardware formula representation, which can result in diverse data flow and timing features of decoding in implementation. Some existing architectures [15], [17], [37], [38] for LDPC/polar codes can be formulated as Fig. 5. These architectures can be represented in terms of degree and parallelism. The unified representation for different architectures provide the foundation for the unified architecture design of LDPC/polar codes.

The hardware formula representations in Fig. 5 primarily reflect the degree and parallelism of the architectures. However, the specific design of these nodes and the interconnection

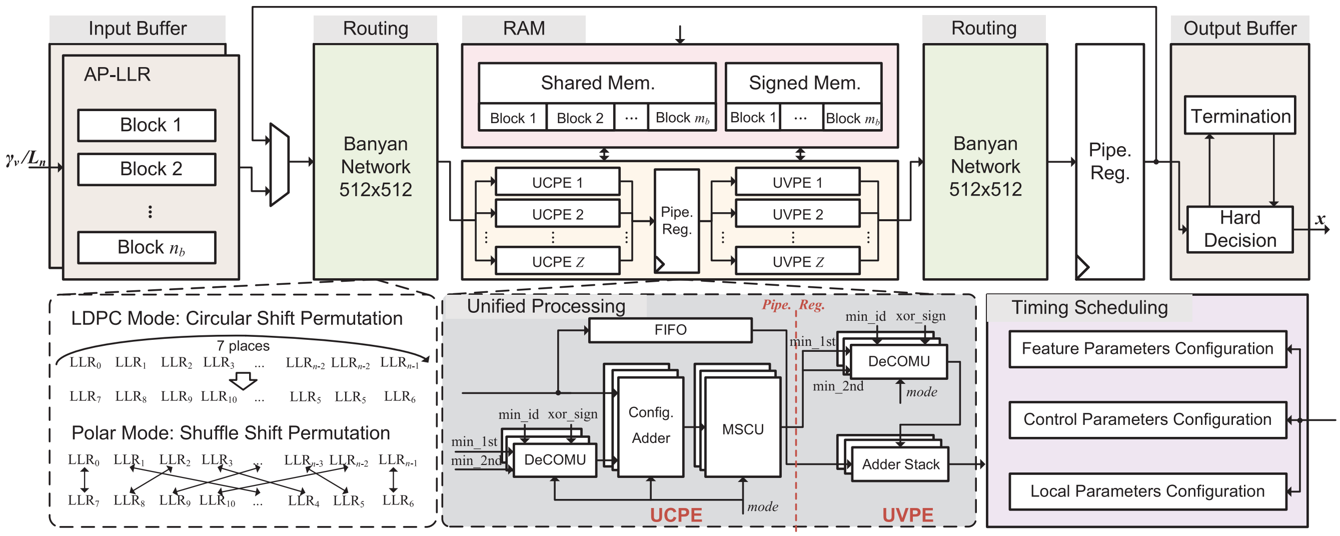


Fig. 6. Overall architecture of the UniDec.

between them cannot be adequately depicted in Fig. 5. For example, the notation "$\mathrm{CMU}_d^{L/P}$" only denotes the degree-d CN function module of LDPC/polar codes but the specific design details are not provided. Therefore, formula representation only reflects the architecture with different design parameters, such as degree and parallelism. Additionally, the detailed design of unified processing nodes and configurable permutation networks will be discussed in the following section.

IV. Unified Architecture for 5G NR LDPC/Polar Codes

A. Overall Architecture

Fig. 6 shows the overall architecture of the proposed UniDec. The UniDec can be divided into three main parts: 1) unified processing nodes, 2) configurable permutation networks, and 3) extrinsic memory units. The unified processing nodes and configurable permutation networks, regarded as the core decoding logic units, can be reconfigured to decode under two decoding modes with multi-frame processing. The extrinsic memory units consist of data information memory and control information memory, utilized for storing intermediate messages and managing timing scheduling, respectively.

In the overall architecture, we should confirm its degree and parallelism parameters as the hardware formula representation in Fig. 5. Regarding degree selection, we reuse the low-degree nodes to achieve high-degree functions, ensuring high flexibility and low hardware complexity. As for parallelism selection, it is important to consider the factor graph structures for both codes and aim to balance throughput and hardware utilization. Therefore, we adopt block parallel layer decoding for LDPC codes and single column decoding for polar codes in this paper. With the maximum lifting value $Z_{\max} = 384$ and polar code length $N_{\max} = 2^{n_{\max}} = 512$, we determine the node parallelism P of unified processing nodes as follows.

$$P = \max\left(Z_{\max}, \frac{N_{\max}}{2}\right). \tag{8}$$

To maximize utilization for the given node parallelism P, we employ multi-frame processing to enhance hardware efficiency. The detailed design and multi-parallelism configuration are presented in the following sections.

B. Unified Processing Nodes

The unified processing node consists of two parts: the unified CN processing element (UCPE) and the unified VN processing element (UVPE). Further subdivision of these two parts results in five components, including data decompression units (DeCOMU), configurable adders, min-sum computation units (MSCU), first-in-first-out memory (FIFO), and adder stacks. Given the theoretical foundation outlined in (1)–(6), we delve into the detailed node design to optimize the common operational logic for both codes efficiently. Depended on the signal *mode* configuration, the decoding architecture can switch between LDPC mode and polar mode, with *mode* 0 for LDPC codes and the *mode* 1 for polar codes. The detailed designs are listed as follows.

1) DeCOMU: The DeCOMU aims to restore the messages computed in the previous iteration of CN processing. In the LDPC mode, the $\alpha_{c,v}$ message restoration requires the two most minimums (min_1st, min_2nd) of $\beta_{v',c}^{(t-1)}$, $v' \in \mathcal{V}_c$, the index of min_1st (min_id), the sign bits of $\beta_{v',c}^{(t-1)}$ for each VN v' and their XORed sign (xor_sign). The above messages are updated and stored in the shared memory and signed memory during each iteration. In CN processing, min_1st and min_2nd from previous iteration are used as the input1 and input2 for the DeCOMU in Fig. 7, min_id, xor_sign, and the sign bits of $\beta_{v',c}^{(t-1)}$ for each VN v' are also provided to DeCOMU to restore $\alpha_{c,v'}^{(t-1)}$ as (4). After passing through sign-magnitude to 2's compliment (STC) units, the $\alpha_{c,v'}^{(t-1)}$ message can be input to the configurable adder for $\beta_{v',c}^{(t)}$ updating as (1).

In polar decoding, updating stage messages in a given direction involve corresponding stage messages from the opposite direction in the previous iteration. Therefore, the $R_{k,2j}/R_{k,2j+1}$ ($L_{k+1,j}/L_{k+1,j+N/2}$) are supplied as input1 and input2 to

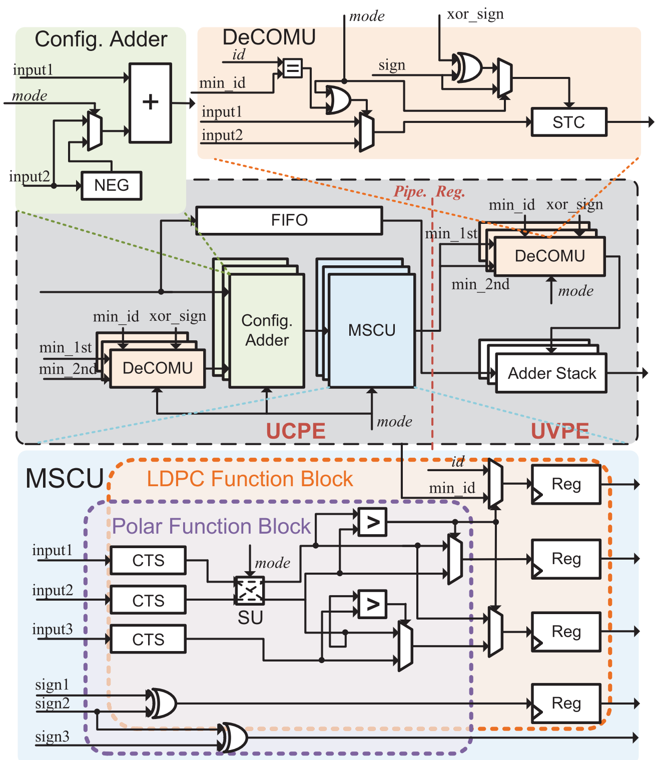


Fig. 7. Unified processing node design in the UniDec.

the DeCOMU when updating stage messages during $\boldsymbol{L}/\boldsymbol{R}$ propagations. It should be noted that the min_id is required to select between input1 and input2 to satisfy $v' \in \mathcal{V}_c \backslash v$ in (4) in the LDPC mode. In the polar mode, the DeCOMU only selects $R_{k,\,2j+1}$ $(L_{k+1,\,j+N/2})$ by the signal *mode* to achieve iterative computation as (5).

2) Configurable Adder: The configurable adder is tasked with processing extrinsic message during VN updates. In the LDPC mode, the configurable adder receives AP-LLR message $\boldsymbol{\gamma}^{(t)}$ and $\boldsymbol{\alpha}^{(t-1)}$ message, and the extrinsic message $\beta^{(t)}$ is computed within the configurable adder which performs as a subtractor for VN updating as (1). In the polar mode, the configurable adder executes $R_{k,\,2j+1} + L_{k+1,\,j+N/2}$ and then proceeds to achieve the first and third equations in (5) at the MSCU module.

3) MSCU: The MSCU serves as the core of the UCPE for CN updating. The MSCU is composed of 2's compliment to sign-magnitude unit (CTS), comparing unit, and switch unit (SU). The SU has two states referred to as 'BAR' and 'CROSS' when the signal *mode* equals '0' and '1'. Fig. 7 displays the detailed architecture of the MSCU, with parts colored in orange and purple to differentiate the functions related to LDPC and polar decoding modes, respectively.

In the LDPC mode, the MSCU is responsible for generating the compressed CN messages including min_1st, min_2nd, min_id, xor_sign, and the sign bit of the $\beta^{(t)}_{v'c}$ messages of VNs v' connected to CN c, $v' \in \mathcal{V}_c$. Due to the block-parallel decoding architecture used in the LDPC mode, the compressed CN message is obtained by reusing the MSCU d_c times. Therefore, the current magnitude of $\beta^{(t)}$ message, serving as input2, is compared with the last min_1st and min_2nd (input1 and input3) to achieve CN updating as (4) when the SU is in the 'BAR' state. In addition to min_1st and min_2nd, the registers in Fig. 7 are also used to update xor_sign and min_id for the current CN c.

In the polar mode, the single-column architecture can update the messages for one stage in a single step. Hence, the incoming data $L_{k+1,j}$, $L_{k+1,j+N/2} + R_{2j+1}$, and $R_{k,2j}$, corresponding to the $\phi(a,\ b+c)$ and $\phi(a,\ d)$ function in (5), are selected as input1, input2, and input3 for the MSCU module. The module updates $L_{k,i}$ and $R_{k+1,i}$ in (5) when the SU is in the 'CROSS' state.

4) FIFO: In the block parallel scheduling of LDPC mode, the MSCU requires d_c clock cycles to compute the compressed CN messages. After that, the AP updating will be executed for all d_c VNs connected to the CN c as (3). To decouple the AP-LLR updating from CN updating, a FIFO, only works in the LDPC mode, is utilized to temporarily store the compressed CN messages for $\gamma^{(t)}$ message updating. The bit width of the FIFO is determined by the bit width of the compressed CN messages and the maximum lifting values Z, while the depth of the FIFO depends on the maximum row weight of the base graphs (BG1 and BG2) in 5G NR standard.

5) Adder Stack: The adder stack is designed to handle the message output from the MSCU in this iteration. In the LDPC mode, the adder stack performs the updating of $\gamma^{(t)}$ messages as specified in (3). In the polar mode, the output from MSCU needs to be added $L_{k+1,j+N/2}/R_{k,2j+1}$ to achieve the updating of $L_{k,2j+1}/R_{k+1,j+N/2}$ as described in the second/fourth equations in (5).

Fig. 8 provides an overview of the UCPE/UVPE design and visualizes hardware activation in both LDPC and polar modes. To differentiate between the structures and connections in each mode, the distinctive elements of polar codes are highlighted in blue in Fig. 8(b), while the non-essential components in the polar mode are shaded in grey when compared to the LDPC mode in Fig. 8(a).

C. Configurable Permutation Networks

The configurable permutation network is a crucial component for enhancing the flexibility and parallelism in the UniDec. The Banyan network [39] can be an excellent choice to establish a reconfigurable connection. The structure of a $P_N \times P_N$ combined Banyan network consists of two $P_N \times P_N$ Banyan networks with a total of $p_n + 1$ stages, where $p_n = \log_2 P_N$ and P_N represents the network scale. In this paper, we adopt a 512×512 combined Banyan network with 10 stages, including 9 stages of two 512×512 Banyan networks and 1 interMUX stage.

The recursive structure of the Banyan network generated by SUs is depicted in Fig. 9. In this section, we propose MUX control schemes for both codes to facilitate unified processing node interconnection. Furthermore, a multi-parallelism strategy is also introduced for multi-frame processing.

1) Circular Shift Permutation for LDPC Mode: The Banyan networks are utilized in LDPC codes and enable the circular shift values SV or input data within the network scale, where

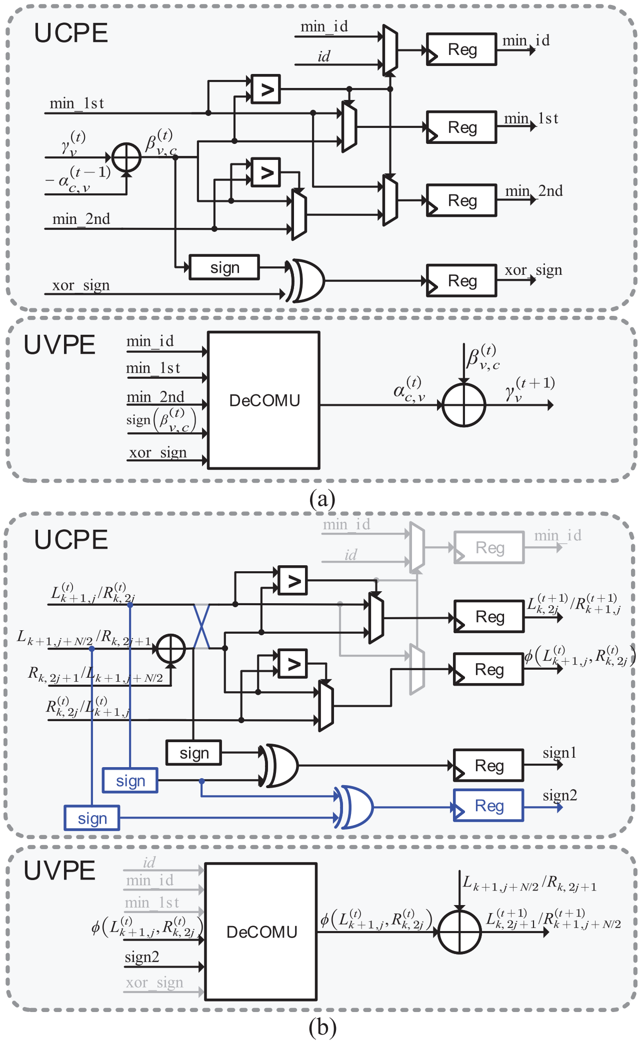


Fig. 8. Overview of UCPE and UVPE designs in (a) LDPC mode and (b) polar mode.

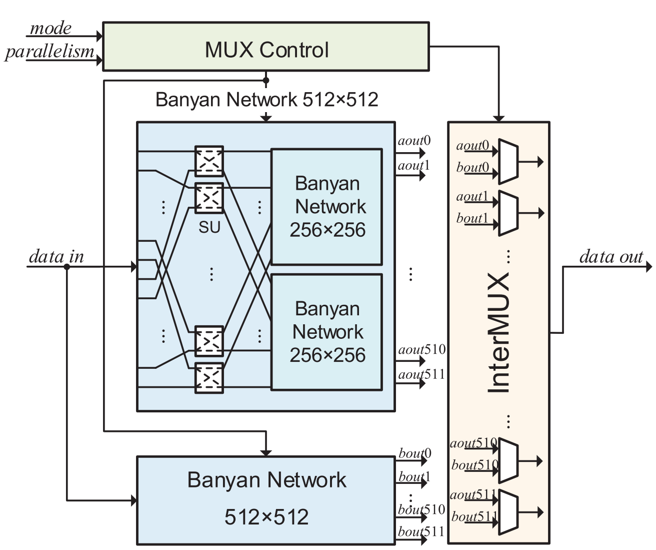


Fig. 9. 512 × 512 combined Banyan network in the UniDec.

$0 \le SV \le P_N$. However, they still face numerous challenges in practical applications.

In [39], extensive modulo operations increase the decoding delay of the implementation. The mapping scheme [40] eliminates all modulo operations to shorten the critical path. However, this approach requires pre-storing numerous control signals for multi-parallelism. Therefore, we introduce a control signal scheme to simplify its implementation. Given that the Banyan network scales with a power-of-two, our proposed control signal scheme eliminates all modulo operations [39] and saves pre-stored memory for control signals [40].

We divide the simplified control signal scheme into two parts: the first p_n Banyan network stages and the final interMUX stage. Initially, the $(k+1)$-th stage of the control signals C_k for Banyan network, $0 \le k < p_n$, can be summarized as

$$C_k = \begin{cases} \text{zeros}(l_k - s_k, 1) \boxplus \text{ones}(s_k, 1), & s_k \le l_k, \\ \text{zeros}(2l_k - s_k, 1) \boxplus \text{ones}(s_k - l_k, 1), & \text{otherwise}, \end{cases} \tag{9}$$

where the functions zeros$(a, 1)$ and ones$(a, 1)$ denote a $1 \times a$ zero vector and one vector, respectively, and the operator $\boxplus$ is a splicing operation, e.g., $a \boxplus a = [a\ a]$. In (9), the parameters s_k and l_k, which generate control signals C_k for $(k+1)$-th stage, are determined by the circular shift value SV and the network scale P_N, as defined below:

$$\begin{cases} s_k = & \text{bin}(s)[p_n - k - 1 : 0], \quad s = SV, \\ l_k = & \text{bin}(l) \gg k, \quad\quad\quad\quad\; l = \frac{P_N}{2}, \end{cases} \tag{10}$$

where the operators $[k-1:0]$ and $\gg k$ represent the least significant k bit interception and k bit right shift for the binary form of the variable a, i.e., bin(a), respectively. The intercept and shift operations in (10) can significantly simplify computations, allowing all control signals of p_n stages to be generated simultaneously. This not only shortens the critical path but also mitigates hardware implementation complexity.

Benefited from the recursive structure, the Banyan network can be divided into multiple subnetworks to achieve different levels of parallelism. The parallelism of the combined Banyan network is defined as $M_P = \log_2\left(\frac{P_N}{P_M}\right)$, where P_M is the required scale of the Banyan subnetwork after splitting. And it is evident that the control signals C_k determined as (9)(10) is a special case when $P_M = P_N$. In contrast, when the parallelism equals to k', $k' > 0$, the control signals of the first k' stage are set to zero, e.g., $C_k = \text{zeros}\left(\frac{P_N}{2}, 1\right)$, $0 \le k < k'$.

For the final interMUX stage, the $p_n + 1$-th stage of the control signal C_{p_n} is

$$\begin{cases} C_{p_n} = (2^{M_P}) \boxtimes C'_{p_n}, \\ C'_{p_n} = \text{ones}(SV, 1) \boxplus \text{zeros}(P_M - SV, 1), \end{cases} \tag{11}$$

where the operator $\boxtimes$ is a multiple splicing operation, e.g., $3 \boxtimes a = a \boxplus a \boxplus a$.

The equations of (9) and (11) determine all the control signals for each stage based on the parameters P_N, P_M, and SV. Besides, It should be noted that for the scheme mentioned above, the configuration of shift values for the top and bottom Banyan networks depends on the parallelism of the Banyan network and lifting values. Considering the parameters P_M and Z, the above scheme assigns shift values SV_1 for the top Banyan network, where $SV_1 = SV$. As for the shift value SV_2 of the bottom Banyan network, it must satisfy the condition

$SV_2 = P_M - Z + SV_1$. At this point, the arbitrary shift values and input data size within the scale P_N can be achieved.

2) Shuffle Shift Permutation for Polar Mode: Compared with the application of Banyan networks in LDPC decoding, there is currently no control scheme aimed at enhancing the flexibility of polar decoders using Banyan networks. The fixed connections in the BP decoder [41] are designed for a single code length without flexibility. To enhance the flexibility, a half column architecture is proposed in [42] to accommodate code lengths. However, the stage messages need to be divided into multiple memory banks and processed with multiple clocks. This results in a reduction in throughput and hardware efficiency. Therefore, we propose the first multi-length/parallelism control signal scheme for polar codes based on the Banyan network to improve the throughput across different code length scenarios.

Based on the factor graph with shuffle shift permutation [41] for implementation, the shuffle/reverse-shuffle operators for $\boldsymbol{L}/\boldsymbol{R}$ propagations between stages are depicted in Fig. 2(b). The operators can be denoted as the transformation functions $\mathtt{L}(i)$ and $\mathtt{R}(i)$, $0 \le i < N$.

$$\mathtt{L}(i) \rightarrow \begin{cases} 2i, & 0 \le i < \frac{N}{2}, \\ 2\left(i - \frac{N}{2}\right) + 1, & \text{otherwise}. \end{cases}$$
$$\mathtt{R}(i) \rightarrow \begin{cases} \frac{i}{2}, & i \,\%\, 2 = 0, \\ \frac{i}{2} + \frac{N}{2}, & \text{otherwise}. \end{cases} \tag{12}$$

Additionally, with the introduction of 512×512 Banyan network, we can leverage the parallelism of the Banyan network to facilitate multi-code parallel decoding, thus improving the hardware implementation efficiency.

The control signals C^{top} and C^{bot} for the top and bottom Banyan network are discussed for $\boldsymbol{L}/\boldsymbol{R}$ propagations, respectively. To achieve the function $\mathtt{L}(i)$, the control signals C_k^{top} for the first p_n stage, $0 \le k < p_n$, can be expressed as

$$\begin{cases} C_0^{\text{top}} = \text{zeros}\left(\frac{P_N}{4}, 1\right) \boxplus \text{ones}\left(\frac{P_N}{4}, 1\right), \\ C_{p_n-1}^{\text{top}} = \left(\frac{P_N}{4}\right) \boxtimes (\text{zeros}(1,1) \boxplus \text{ones}(1,1)), \\ C_k^{\text{top}} = (2^{k-1}) \boxtimes C_k^{\text{cell}}, \quad 1 \le k < p_n - 1, \end{cases}$$
$$\text{where} \begin{cases} C_k^{\text{cell}} = C'_k \boxplus \sim C'_k, \\ C'_k = \text{zeros}(2^{p_n-k-2}, 1) \boxplus \text{ones}(2^{p_n-k-2}, 1). \end{cases} \tag{13}$$

And the bottom control signals C_k^{bot}, $0 \le k < p_n$, are

$$C_k^{\text{bot}} = \begin{cases} C_k^{\text{top}}, & 1 \le k < p_n - 1, \\ \sim C_k^{\text{top}}, & \text{otherwise}, \end{cases} \tag{14}$$

where the operator $\sim$ in (13)(14) is a bitwise invert operation. Finally, the control signal C_{p_n} of the final interMUX stage can be denoted as

$$C_{p_n} = \left(\frac{P_N}{2}\right) \boxtimes (\text{zeros}(1,1) \boxplus \text{ones}(1,1)). \tag{15}$$

According to the equations of (13), (14), and (15), the control signals for each stage can be determined for $\boldsymbol{L}$ propagation. For the parallelism configuration for $\boldsymbol{L}$ propagation, the control signals $C_k^{\text{top/bot}}$, $0 \le k < k'$, are set to zero when the parallelism

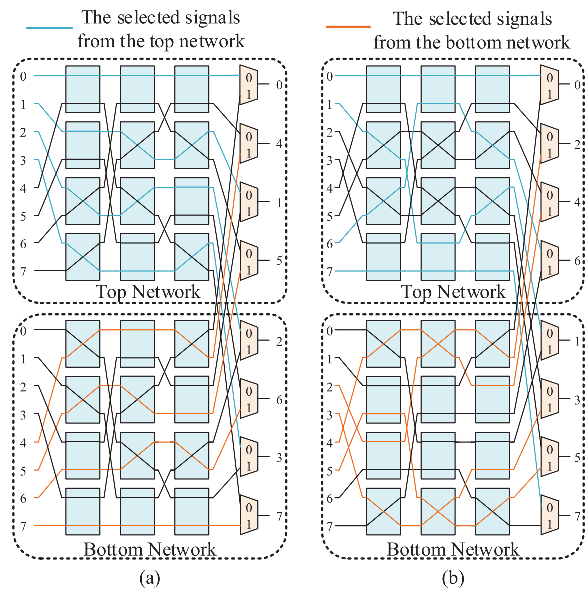


Fig. 10. (a) $\boldsymbol{L}$ and (b) $\boldsymbol{R}$ propagations of polar codes for combined Banyan network, $P_N = 8$, $M_P = 0$.

M_P equals to k', $k' > 0$, and the $k'+1$-th stage of the control signal should be reconfigured as

$$C_{k'}^{\text{top}} = \sim C_{k'}^{\text{bot}} = (2^{k'}) \boxtimes C'_{k'}. \tag{16}$$

The aforementioned control signal scheme can achieve the $\boldsymbol{L}$ propagation with multi-parallelism for polar codes. For $\boldsymbol{R}$ propagation, we present the control scheme generated through a recursive process as follows.

$$C_k^{\text{top}} = \sim C_k^{\text{bot}} = C_{\text{rec.}}^{(p_n-1)}, \ 0 \le k < p_n, \text{ where}$$
$$\begin{cases} C_{\text{rec.}}^{(t)} = C_{\text{rec.}}^{(t-1)} \boxplus \sim C_{\text{rec.}}^{(t-1)}, \ 1 \le t < p_n, \\ C_{\text{rec.}}^{(0)} = \text{zeros}(1,1). \end{cases} \tag{17}$$

Similar to the parallelism configuration for the $\boldsymbol{L}$ propagation, the $\boldsymbol{R}$ propagation control signals for the first k' stages are set to zeros when the parallelism M_P equals to k', $k' > 0$. Then the remaining stages should be reconfigured as

$$C_k^{\text{top}} = \sim C_k^{\text{bot}} = (2^{k'}) \boxtimes C_{\text{rec.}}^{(p_m-1)}, \ k' \le k < p_n, \tag{18}$$

where $p_m = \log_2 P_M$. Moreover, the control signal for the final interMUX stage is also influenced by the Banyan network's parallelism. The control signal for the $\boldsymbol{R}$ propagation in the p_n+1-th stage is as follows:

$$C_{p_n} = (2^{p_n - p_m + 1}) \boxtimes C_{\text{rec.}}^{(p_m-1)}. \tag{19}$$

It is worth noting that when the parallelism M_P is set to 0, the parameters p_m and p_n are equivalent.

In conclusion, we propose the control signals of $\boldsymbol{L}/\boldsymbol{R}$ propagation with multi-parallelism in (13)–(19). For a detailed explanation, we analyze the case of $\boldsymbol{L}$ propagation with $P_N = 8$ as an example, the control signals in Fig. 10(a) are

$$\begin{aligned} &C_0^{\text{top}} = [0\ 0\ 1\ 1],\ C_1^{\text{top}} = [0\ 1\ 1\ 0],\ C_2^{\text{top}} = [0\ 1\ 0\ 1], \\ &C_0^{\text{bot}} = [1\ 1\ 0\ 0],\ C_1^{\text{bot}} = [0\ 1\ 1\ 0],\ C_2^{\text{bot}} = [1\ 0\ 1\ 0], \\ &C_3 = [0\ 1\ 0\ 1\ 0\ 1\ 0\ 1]. \end{aligned} \tag{20}$$

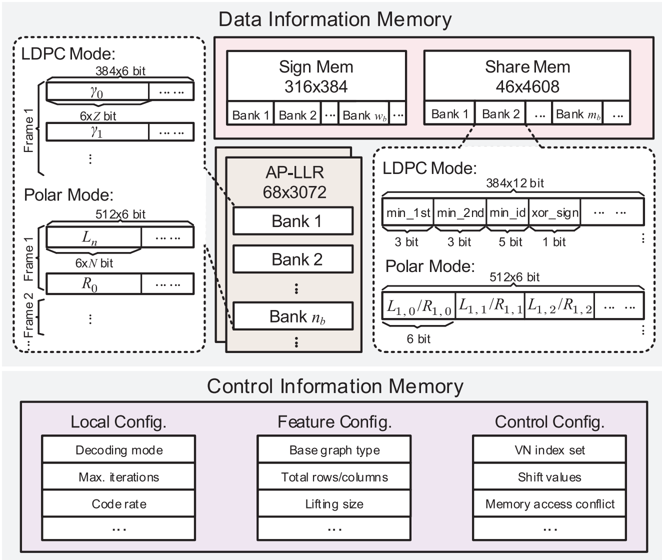


Fig. 11. Memory structure of the UniDec.

Similarly, we give the control signals for $\boldsymbol{R}$ propagation with $P_N = 8$ in Fig. 10(b), denoted as

$$\begin{aligned} &C_0^{\text{top}} = [0\ 1\ 1\ 0],\ C_1^{\text{top}} = [0\ 1\ 1\ 0],\ C_2^{\text{top}} = [0\ 1\ 1\ 0], \\ &C_0^{\text{bot}} = [1\ 0\ 0\ 1],\ C_1^{\text{bot}} = [1\ 0\ 0\ 1],\ C_2^{\text{bot}} = [1\ 0\ 0\ 1], \\ &\ C_3 = [0\ 1\ 1\ 0\ 0\ 1\ 1\ 0]. \end{aligned} \tag{21}$$

D. Extrinsic Memory Units

Fig. 11 illustrates the structure of the extrinsic memory units. The memory sharing structure contains the data information memory and control information memory. The data information memory stores the channel soft message and intermediate message. Meanwhile, the control information memory retains the configurable control instructions for timing management, ensuring compatibility with the 5G NR standard. Additionally, detailed quantization and memory utilization schemes for both decoding modes are also provided in Fig. 11. In the polar mode, the $\boldsymbol{L}/\boldsymbol{R}$ messages are uniformly quantized to $Q_P = 6$ bits including 2 fractional bits [36], while the 6-bit quantization scheme [26] is also adopted for our decoder under the LDPC mode.

1) Data Information Memory: The data information memory can be divided into three parts: the channel AP-LLR memory, the share memory storing the intermediate message produced by unified processing nodes, and the sign memory activated only in the LDPC mode. As shown in Fig. 11, the total memory capacity of the three memories is 542,208 bits, involving 208,896 bits allocated to AP-LLR memory, 211,968 bits to shared memory, and 121,344 bits to signed memory.

Note that the AP-LLR memory keeps the channel probabilistic information, its bit length is determined by the sub-matrix size Z for block parallel LDPC decoding. And the depth is dependent on the number of columns n_b in the base graph. In polar mode, $\boldsymbol{L}_n$ and $\boldsymbol{R}_0$ messages are stored in the memory for each frame. The parameters of AP-LLR memory ($D \times W$) are illustrated in Fig. 11, where D, W denote the bit width and depth of memory, respectively.

The share memory stores the intermediate message of unified processing nodes in each iteration. There is a notable difference in the intermediate message formats between the two codes. In the LDPC mode, the intermediate message consists of min_1st, min_2nd, min_id, and the xor_sign produced by the corresponding CNs. In contrast, the share memory stores the stage messages $\boldsymbol{L}_i/\boldsymbol{R}_j$, $0 < i,\ j < n$ in the polar mode. Finally, the bit width and depth of the share memory are determined as follows.

$$\begin{cases} W = \max(Z_{\max} \cdot Q_L,\ N_{\max} \cdot Q_P), \\ D = \max(n_b,\ n-1), \end{cases}$$

where $Q_L = Q_{\text{min\_1st}} + Q_{\text{min\_2nd}} + Q_{\text{min\_id}} + 1$.

The sign memory is reserved only for LDPC mode to store sign messages derived from the $\beta^{(t)}$ message for each CN. To support the two types of base graphs in 5G NR LDPC codes, the bit width and depth of the sign memory depend on the maximum lifting value $Z_{\max}$ and the maximum number of non-zero elements w_b for BG1/BG2.

In summary, the amount of data that needs to be stored in LDPC or polar mode is consistent with that of dedicated LDPC/polar decoders [15], [38]. In the proposed design, the data memory stores the intermediate data generated in both modes without introducing any additional memory overhead.

2) Control Information Memory: The control information memory is used to store configurable control instructions for decoding scheduling. These instructions are generated by the local parameter, feature parameter, and control parameter configuration. Table I shows the 5G NR feature parameter configuration for both codes. We set up the parameters based on local parameters such as the decoding mode, information bit length K, and code rate R. The control parameters are derived from the feature parameters to execute decoding scheduling, ensuring compatibility with 5G NR LDPC/polar decoding.

According to the feature parameters in Table I, the 5G NR LDPC codes and polar codes can be denoted as $\mathcal{L}(bg, m_b, n_b, Z)$ and $\mathcal{P}(N, \mathcal{A})$, respectively. While $\mathcal{P}(N, \mathcal{A})$

TABLE I
5G NR FEATURE PARAMETERS CONFIGURATION

Input Parameters:	
Decoding mode $mode$, information bit length K, code rate R.	
Feature Parameters:	
LDPC mode ($mode = 0$)	Base graph type bg, $bg \in \{0, 1\}$.
	Total rows of base graph m_b, $\begin{cases} 5 \le m_b \le 46, \text{when } bg = 0, \\ 5 \le m_b \le 42, \text{when } bg = 1. \end{cases}$
	Total columns of base graph n_b, $\begin{cases} 27 \le n_b \le 68, \text{when } bg = 0, \\ 15 \le n_b \le 52, \text{when } bg = 1. \end{cases}$
	Lifting size Z, $Z \in \{Z_0, Z_1, \cdots, Z_{50}\}$.
Polar mode ($mode = 1$)	Code length N, $N = 2^n$, $7 \le n \le 9$.
	Code construction for sub-channel allocation $\mathcal{A}$.

only requires the stage number n as the control parameter to regulate $\boldsymbol{L}/\boldsymbol{R}$ propagation scheduling, $\mathcal{L}(bg, m_b, n_b, Z)$ should consider the VN index set $\mathcal{V}_c$ connected to the CN c, shift values SV in base graphs, and memory access conflict ρ for block parallel layer scheduling. Since memory access conflicts can only occur between adjacent layers of CNs in the base graph, we construct memory conflict set ρ as (22).

$$\rho(i,j) = \begin{cases} 1, & \mathcal{V}_{c(i)}(j) \in \mathcal{V}_{c(i-1)} \cap \mathcal{V}_{c(i)}, \\ 0, & \text{otherwise}, \end{cases} \quad 0 \le j < d_{c(i)}, \tag{22}$$

where $c(i)$ denotes the CNs in the i-layer of the base graph. Then, we execute read access scheduling as the condition (23).

$$\begin{aligned} &\rho(i,j) == 0, \\ &\text{or } \rho(i,j) == 1 \parallel \mathcal{V}_{c(i)}(j) < \mathcal{V}_{c(i-1)}(j'). \end{aligned} \tag{23}$$

where $\mathcal{V}_{c(i)}(j)$, $\mathcal{V}_{c(i-1)}(j')$ represent the current VN indices for read access and write-back, respectively.

The control information memory in Fig. 11 shows feature parameters and control parameters for both modes. And the latency analysis will be presented in the following section.

E. Timing Schedule and Latency Analysis

For LDPC decoding, the block parallel layer scheduling can decouple the data dependency between VNs and CNs. However, the stall clock cycles n_{hazard} caused by read-write conflicts do affect the overall decoding latency. Fortunately, the base graphs for 5G NR LDPC codes exhibit good orthogonality between adjacent rows, which helps mitigate the impact of read-write conflicts on decoding latency.

To improve the hardware efficiency, the proposed multi-frame processing strategy based on the combined Banyan networks allows for the simultaneous decoding of M_F frames, where $M_F = \frac{P}{\lceil Z \rceil_2}$, and $\lceil\ \rceil_2$ represents the ceil operation to the nearest power of 2. Therefore, the throughput of the decoder under LDPC mode (Thp_L) can be computed as

$$\text{Thp}_L = \frac{\text{Freq.} \times n_b \times Z \times M_F}{(w_b + n_{\text{hazard}}) \times T_{\text{iter}}}, \tag{24}$$

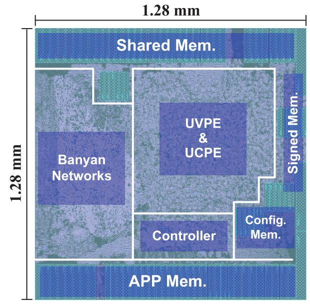
where T_{iter} denotes the number of iterations.

In polar decoding, the single-column architecture consists of a maximum of 256 PEs for one stage. For cases where the code length is less than 512, we configure multiple frames in parallel under the Banyan network, enabling the decoding of M_F frames simultaneously, $M_F = \frac{N_{\max}}{\lceil N \rceil_2}$. Hence, the throughput of the decoder under polar mode (Thp_P) can be computed as

$$\text{Thp}_P = \frac{\text{Freq.} \times N \times M_F}{(n-1) \times T_{\text{iter}}}. \tag{25}$$

V. Implementation Results

This section presents the implementation results of the proposed UniDec, along with a comparison to SOA decoders, including LDPC-only decoders, polar-only decoders, and decoders that support both codes.



Process	40nm CMOS
Code Type	5G NR LDPC/Polar Codes
Power [mW]	135.83
Logic Area [mm^2]	0.56
Mem. Area [mm^2]	0.87
Core Area [mm^2]	1.28 × 1.28

Fig. 12. Post-layout and implementation results of the UniDec.

A. Hardware Implementation Details

Fig. 12 presents the post-layout and implementation results of the UniDec, synthesized by the Synopsys Design Compiler and placed/routed with the Synopsys IC Compiler in 40 nm CMOS technology. The values are quantized to 6-bit fixed-point numbers and the node parallelism P is set to 384. The UniDec is fully compatible with the 5G NR standard and supports multi-frame processing for flexible parallel decoding.

The UniDec occupies a core area of 1.64 mm^2, comprising 0.56 mm^2 of logic and 0.87 mm^2 of memory. Figure 12 highlights the logic components including the UVPE, UCPE, Banyan networks, and controller, as well as the memory components, which consist of AP-LLR memory, shared memory, signed memory, and configuration memory. Based on the parameters in Section IV-D, all memories are implemented using two-port register files. The power dissipation of the decoder is 135.83 mW at 500 MHz clock frequency. It can deliver peak throughputs of 21.87 Gb/s in LDPC mode and 10.24 Gb/s in polar mode, respectively.

The comprehensive design validation has been conducted by configuring control information, including decoding modes and code parameters. For a fair comparison, we refer to the code parameters of SOA decoders to provide several similar decoding cases for performance comparison. The comprehensive analyses are conducted on multi-mode decoders and dedicated decoders under different code parameters.

B. Comparison With SOA Decoders

The comparison results in Table II can be divided into three categories to achieve a comprehensive analysis for the UniDec: LDPC-only decoders, polar-only decoders, and the decoders supporting the two codes.

To demonstrate the advantages of the proposed architecture, we make the comparisons with the SOA multi-mode decoders [26], [31]. According to Table II, our UniDec demonstrates comprehensive superiority in both LDPC and polar modes compared with previous work [26], [31]. Thanks to the unified architecture with high node parallelism, the peak throughput of the UniDec outperforms the SOA decoder [26] with a maximum improvement of 33.64× and 6.06× in LDPC and polar modes, respectively. Meanwhile, the UniDec also attains a maximum of 5.98× and 1.07× area efficiency over [26], respectively. Compared to the multi-mode decoders [31] that only partially support the 5G NR standard, the UniDec offers

TABLE II

HARDWARE RESULTS COMPARISON FOR SOA MULTI-MODE DECODERS AND DEDICATED DECODERS

	This work		[SSC-L'22] [31]		[TVT'21] [26]		[TCASI'24] [21]	[TCASI'22] [20]	[TCASI'23] [22]	[TCASI'19] [17]
Implementation	Post-layout		Silicon		Synthesis		Post-layout	Post-layout	Post-layout	Post-layout
Process [nm]	40		40		55		65	65	65	40
Standard Compatible	5G NR		5G NR*		5G NR†		5G NR	5G NR	5G NR	N/A
Code Type	LDPC	Polar	LDPC	Polar	LDPC	Polar	LDPC	LDPC	Polar	Polar
Algorithm	OMS-BP		MS-BP		MS-BP		NMS-BP	OMS-BP	SSCL-8	MS-BP
Architecture	Block Parallel	Single Column	Partial Parallel	Folded Column	Row Parallel	Folded Column	Partial Parallel	Partial Parallel	Tree Search	Double Column
Multi-frame Process.	Yes		No		No		No	No	No	No
Quantization [bit]	6		6		6		$7, 5$	8	6	–
Code Length	Up to 26112	Up to 512	6400	1024	2720	1024	Up to 26112	Up to 26112	1024	1024
Code Rate	All cases of 5G		–	–	$11/34$	$1/2$	All cases of 5G	All cases of 5G	$1/2$	$1/2$
Core Area [mm^2]	1.64		2.07		0.55		3.82	5.74	3.89	0.70
Max. Freq. [MHz]	500		280		300		500	500	300	500
Power [mW]	136		169		–		359	413	389	423
Avg. Iterations	3	5	5	4.4	5		3		1	7.5
Thp.‖ [Gb/s]	21.87	10.24	2.29	1.85	0.47	1.23	23.47	21.78	1.02	7.61
Scaled to 40 nm, and 1.0 V‡										
Norm. Power [mW]	136		169		–		221	254	239	423
Norm. Thp. [Gb/s]	21.87	10.24	2.29	1.85	0.65	1.69	38.14	35.39	1.66	7.61
Energy [pJ/b]	6.22	13.28	73.80	91.35	–	–	5.79	7.18	234.46	55.58
Area Eff. [Gb/s/mm^2]	13.34	6.24	1.11	0.89	2.23	5.81	23.12	16.31	1.13	10.80

* The decoder only supports LDPC codes based on the 5G BG2 with up to 128 lifting size.
† The configurable codeword parameters of the decoder are limited by the fixed finite types of decoding patterns.
‖ The notation Thp. denotes the peak throughput of the decoders that can support all cases of 5G NR standard.
‡ Frequency $\propto s$, energy $\propto \frac{1}{s}(\frac{1.0}{V_{dd}})^2$ and area $\propto \frac{1}{s^2}$, where s is the scaling factor to 40 nm.

full compatibility with both 5G NR LDPC/polar codes, while improving throughput and area efficiency through multi-frame processing.

Compared with the SOA LDPC-only decoders [20], [21], the proposed decoder still demonstrates competitive performance. In the LDPC mode, the UniDec achieves a 15.43% reduction in energy consumption over [20] while maintaining comparable area efficiency. For polar decoding, our decoder is designed specifically for short codes in 5G control channels, and it can deliver a peak throughput of 10.24 Gb/s within the 5G NR standard. In comparison to the SOA polar-only decoder [17], our decoder achieve a 1.35× boost in throughput.

VI. CONCLUSION

This paper presents the UniDec for LDPC/polar codes to meet the requirements of 5G eMBB scenarios. By utilizing the graph theory and hardware formula representation, we propose a unified integration methodology to derive the factor-graph based architecture from graph-theoretic algorithms to hardware implementations. Besides, the unified processing nodes and configurable networks with multi-parallelism are proposed to improve the hardware efficiency. Implemented in 40 nm CMOS technology, the UniDec occupies a size of 1.64 mm^2 core area at 500 MHz clock frequency. Thanks to the unified architecture and its efficient node and network connection designs, the UniDec in LDPC/polar modes outperforms the SOA multi-mode decoder in terms of throughput by 33.64× and 6.06×, and area efficiency by 5.98× and 1.07×, respectively.

Moreover, a joint-graph architecture that includes detection and decoding is part of our future research. This aims to generalize our methodology to other factor-graph based signal processing algorithms.

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